

Joshua Chung

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Mechanical Engineer | Entry Level | Systems Engineer

EDUCATION

University of California, Riverside

BS in Mechanical Engineering

Riverside, CA

June 2025

Certifications: Lean Six Sigma Green Belt

SKILLS

Design/Analysis: Computer-Aided Design (CAD) | Solidworks Simulation | Finite Element Analysis (FEA) |
| Additive Manufacturing | 3D Printing | BOM

Applications: Solidworks | Onshape | Microsoft Excel | Microsoft Word | Microsoft PowerPoint

ENGINEERING EXPERIENCE

NASA Micro-g NExT Capstone

Engineering Project Contributor

University of California, Riverside

Sep 2024 - Mar 2025

- Designed a lunar payload subsystem device using Solidworks integrating multiple mechanical subsystems to collect and store 1-5 mm of lunar regolith resulting in a lightweight and ergonomically suitable for astronaut use
- Developed a cam-based mechanical system and manually operated digging mechanism using Solidworks Simulation ensuring EVA glove compatibility for efficient sample collection
- Implemented manufacturing and assembly processes with 3D printing and CNC machining producing 3+ prototype iterations allowing for easy part replacement while aligning with NASA interface constraints
- Performed design verification and testing using FEA and measurement tools validating the device against strict specifications for safe operation in low-gravity environments
- Created documentation and decision matrices, detailing the trade-offs, performance metrics, and clean sheet design process to support subsystem integration

Bridge Capstone

Lead Simulation Engineer

University of California, Riverside

Apr 2025 - Jun 2025

- Led structural analysis and FEA on a truss bridge to identify stress concentrations and predict failure sites under distributed loads
- Collaborated with a team to design, fabricate, and test 3+ bridge iterations, refining truss geometry through design validation and continuous improvement
- Supported validation design efforts by incrementally applying loads in a controlled test setup documenting structural response and analyzing failure behavior
- Engineered a bridge supporting 5240 grams of load, achieving a structural efficiency ratio of 276 through weight reduction and simulation driven design

Mechanical Design & Assembly Evaluation

Design Engineer

Remote

Feb 2026

- Developed engineering drawings for a machined aluminum component applying callouts and GD&T to ensure manufacturability
- Created structured assembly procedure and bill of materials for a multi-component electromechanical extension system

AI Robotics Laboratory

Research Assistant

University of California Riverside

Apr 2025 - Jun 2025

- Collaborated on a 3-finger robotic gripper for the Ender-3 S1 Pro, researching low-cost gripper designs and refining it through additive manufacturing to enhance robotic manipulation
- Processed 100+ robotic manipulation frames using computer vision tools like Roboflow, cutting preparation time by 20%